Formulario de examen · Estadística Matemática

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Parametrizaciones: el Formulario 2 usa β como **escala** ($=\mathbb{E}$) en Exp/Gamma. La "exponencial de parámetro λ" lleva $\lambda=$ **tasa**: $\beta=1/\lambda$, media $1/\lambda$. $N(\mu,\sigma^2)$ lleva la **varianza**; Geo/Bneg cuentan **ensayos**.

Modelos (parametrización del Formulario 2; si son independientes las FGM se multiplican)

Modelo	$\mathbb{P}(X=x)$ **o** $f_X(x)$	**FDA** $F_X(x)$ $\lfloor\cdot\rfloor$ ent.; I_x beta; P gamma	$\mathbb{E}(X)$	$\mathrm{Var}(X)$	$M_X(t)$
$\mathrm{Ber}(p)$	$p^x(1-p)^{1-x},\ x\in\{0,1\}$	$0\ (x<0);\ 1-p\ (0\le x<1);\ 1\ (x\ge1)$	p	$p(1-p)$	pe^t+1-p
$\mathrm{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x},\ x=0,\dots,n$	$\sum_{k=0}^{\lfloor x\rfloor}\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	$(pe^t+1-p)^n$
$\mathrm{Geo}(p)$	$p(1-p)^{x-1},\ x=1,2,\dots$	$1-(1-p)^{\lfloor x\rfloor}\ (x\ge1);\ 0\ (x<1)$	$1/p$	$(1-p)/p^2$	$\dfrac{pe^t}{1-(1-p)e^t}$
$\mathrm{Bneg}(r,p)$	$\binom{x-1}{r-1}p^r(1-p)^{x-r},\ x\ge r$	$\sum_{k=r}^{\lfloor x\rfloor}\binom{k-1}{r-1}p^r(1-p)^{k-r}$	r/p	$r(1-p)/p^2$	$\Big[\dfrac{pe^t}{1-(1-p)e^t}\Big]^r$
$\mathrm{Hiper}(N,a,n)$	$\binom{a}{x}\binom{N-a}{n-x}/\binom{N}{n}$	$\sum_{k=0}^{\lfloor x\rfloor}\dfrac{\binom{a}{k}\binom{N-a}{n-k}}{\binom{N}{n}}$	$n\frac{a}{N}$	$n\frac{a}{N}(1-\frac{a}{N})\frac{N-n}{N-1}$	sin forma cerrada
$\mathrm{Pois}(\lambda)$	$e^{-\lambda}\lambda^x/x!,\ x=0,1,\dots$	$e^{-\lambda}\sum_{k=0}^{\lfloor x\rfloor}\lambda^k/k!$	λ	λ	$e^{\lambda(e^t-1)}$
Unif. discreta	$1/k,\ x\in\{x_1,\dots,x_k\}$	$\frac{1}{k}\#\{i:x_i\le x\}$	$\frac{1}{k}\sum x_i$	$\frac{1}{k}\sum x_i^2-\bar{x}^2$	$\frac{1}{k}\sum_i e^{tx_i}$
$U(a,b)$	$\frac{1}{b-a},\ a<x<b$	$0\ (x<a);\ \frac{x-a}{b-a}\ (a\le x<b);\ 1\ (x\ge b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\dfrac{e^{tb}-e^{ta}}{t(b-a)}$
$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/(2\sigma^2)}$	$\Phi(\frac{x-\mu}{\sigma})$	μ	σ^2	$e^{\mu t+\sigma^2t^2/2}$
$\mathrm{Exp}(\beta)$	$\frac{1}{\beta}e^{-x/\beta},\ x>0$	$1-e^{-x/\beta}\ (x\ge0);\ 0\ (x<0)$	β	β^2	$(1-\beta t)^{-1}$
$G(\alpha,\beta)$	$\frac{1}{\beta^\alpha\Gamma(\alpha)}x^{\alpha-1}e^{-x/\beta},\ x>0$	$P(\alpha,\frac{x}{\beta})$	$\alpha\beta$	$\alpha\beta^2$	$(1-\beta t)^{-\alpha}$
$\chi^2_{(\nu)}=G(\frac{\nu}{2},2)$	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)}x^{\nu/2-1}e^{-x/2}$	$P(\frac{\nu}{2},\frac{x}{2})$	ν	2ν	$(1-2t)^{-\nu/2}$
$\mathrm{Beta}(\alpha,\beta)$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)},\ 0<x<1$	$I_x(\alpha,\beta)$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\mathbb{E}(X^k)=\frac{B(\alpha+k,\beta)}{B(\alpha,\beta)}$

Dominios FGM: Geo/Bneg: $t<-\ln(1-p)$; Exp/Gamma: $t<1/\beta$; χ^2: $t<\frac{1}{2}$; U: $t\neq0$.

Relaciones (por FGM o definición): Ber_n i.i.d. $\to\mathrm{Bin}(n,p)$; Geo=Bneg$(1,p)$; Bneg=suma de r Geo i.i.d.; $\mathrm{Pois}(\lambda_i)$ indep. $\to\mathrm{Pois}(\sum\lambda_i)$; $\mathrm{Exp}(\beta)=G(1,\beta)$; n Exp(β) i.i.d. $\to G(n,\beta)$; $G(\alpha_i,\beta)$ (mismo β) $\to G(\sum\alpha_i,\beta)$; $N(\mu_i,\sigma_i^2)$ indep. $\to N(\sum\mu_i,\sum\sigma_i^2)$; $aX+b\sim N(a\mu+b,a^2\sigma^2)$; $Z^2\sim\chi^2_{(1)}$; $\sum^\nu\chi^2_{(1)}\to\chi^2_{(\nu)}$; $\chi^2_{(2)}=$Exp(2); $\mathrm{Beta}(1,1)=U(0,1)$.

1. Variable aleatoria, densidad y acumulada

Discreta: $p_X(x)=\mathbb{P}(X=x)\ge0$, $\sum_{S_X}p_X=1$, $\mathbb{P}(X\in A)=\sum_A p_X$.

Continua: $f_X\ge0$, $\int_{-\infty}^{\infty}f_X=1$, $\mathbb{P}(a<X\le b)=\int_a^b f_X$.

$$F_X(x)=\mathbb{P}(X\le x),\quad \mathbb{P}(a<X\le b)=F_X(b)-F_X(a).$$

Continua: $F_X(x)=\int_{-\infty}^{x}f_X(t)\,dt$, $f_X=F_X'$. Discreta: $\mathbb{P}(X=x)=F(x)-F(x^-)$. **Cuantil** p**:** x_p con $F_X(x_p)=p$ (mediana $x_{0.5}$); si F creciente, $x_p=F^{-1}(p)$. **Ojo:** f_X *no* es probabilidad (puede pasar de 1); "$<$" vs "$\le$" da igual en continuas, no en discretas.

2. Esperanza, varianza, covarianza

$$\mathbb{E}(X)=\sum_x x\,\mathbb{P}(X=x)=\int x f_X(x)\,dx,$$

$$\mathbb{E}[g(X)]=\sum_x g(x)\mathbb{P}(X=x)=\int g(x)f_X(x)\,dx.$$

$$\mathrm{Var}(X)=\mathbb{E}\big[(X-\mathbb{E}X)^2\big]=\mathbb{E}(X^2)-[\mathbb{E}X]^2,\quad \sigma_X=\sqrt{\mathrm{Var}(X)}.$$

$$\mathrm{Cov}(X,Y)=\mathbb{E}(XY)-\mathbb{E}(X)\mathbb{E}(Y)=\mathbb{E}\big[(X-\mathbb{E}X)(Y-\mathbb{E}Y)\big].$$

- **E:** $\mathbb{E}(aX+b)=a\mathbb{E}X+b$; $\mathbb{E}(X+Y)=\mathbb{E}X+\mathbb{E}Y$ (siempre); $\mathbb{E}[\sum a_iX_i+b]=\sum a_i\mathbb{E}X_i+b$; $\mathbb{E}(c)=c$.
- $X\perp\!\!\!\perp Y\Rightarrow\mathbb{E}(XY)=\mathbb{E}X\,\mathbb{E}Y$ y $\mathbb{E}[g(X)h(Y)]=\mathbb{E}g(X)\mathbb{E}h(Y)$.
- $\mathbb{E}[g(X)]\neq g(\mathbb{E}X)$ salvo g lineal; g convexa $\Rightarrow\mathbb{E}g(X)\ge g(\mathbb{E}X)$ (Jensen). **Esperanza total:** $\mathbb{E}X=\mathbb{E}[\mathbb{E}(X\mid Y)]$.
- **Var:** $\mathrm{Var}(aX+b)=a^2\,\mathrm{Var}\,X$; $\mathrm{Var}(X+b)=\mathrm{Var}\,X$; $\mathrm{Var}\,c=0$.
- $\mathrm{Var}(X\pm Y)=\mathrm{Var}\,X+\mathrm{Var}\,Y\pm2\,\mathrm{Cov}(X,Y)$; $\mathrm{Var}(\sum X_i)=\sum\mathrm{Var}\,X_i+2\sum_{i<j}\mathrm{Cov}(X_i,X_j)$; si indep. $\mathrm{Var}(\sum X_i)=\sum\mathrm{Var}\,X_i$.
- **Varianza total:** $\mathrm{Var}\,X=\mathbb{E}[\mathrm{Var}(X\mid Y)]+\mathrm{Var}(\mathbb{E}[X\mid Y])$.
- **Cov:** $\mathrm{Cov}(X,Y)=\mathrm{Cov}(Y,X)$; $\mathrm{Cov}(X,X)=\mathrm{Var}\,X$; $\mathrm{Cov}(aX+b,cY+d)=ac\,\mathrm{Cov}(X,Y)$; $\mathrm{Cov}(X+Y,Z)=\mathrm{Cov}(X,Z)+\mathrm{Cov}(Y,Z)$; $\mathrm{Cov}(\sum a_iX_i,\sum b_jY_j)=\sum_i\sum_j a_ib_j\,\mathrm{Cov}(X_i,Y_j)$.
- $X\perp\!\!\!\perp Y\Rightarrow\mathrm{Cov}=0$ (recíproco sólo en la normal bivariada). $\rho=\mathrm{Corr}=\frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}\,X\,\mathrm{Var}\,Y}}\in[-1,1]$, $\mathrm{Corr}(aX+b,cY+d)=\mathrm{sign}(ac)\rho$.
- Cauchy–Schwarz $[\mathrm{Cov}(X,Y)]^2\le\mathrm{Var}\,X\,\mathrm{Var}\,Y$. **Markov** ($X\ge0$): $\mathbb{P}(X\ge a)\le\mathbb{E}X/a$.
- **Chebyshev:** $\mathbb{P}(|X-\mu|\ge k)\le\sigma^2/k^2$; en $k\sigma$: $\le1/k^2$.

3. Función generadora de momentos (FGM)

$$M_X(t)=\mathbb{E}(e^{tX}),\quad M_X(0)=1,$$

$$M_X^{(k)}(0)=\mathbb{E}(X^k),\quad \mathrm{Var}\,X=M_X''(0)-[M_X'(0)]^2.$$

- $M_{aX+b}(t)=e^{bt}M_X(at)$.
- $X_i\perp\!\!\!\perp\Rightarrow M_{\sum X_i}(t)=\prod_i M_{X_i}(t)$; i.i.d. $\Rightarrow[M_X(t)]^n$.
- **Unicidad:** si $M_X=M_Y$ cerca de 0, misma distribución. *Clave del método FGM.*

Receta: FGM de cada sumando (tabla) $\to$ multiplicar (usa M_{a+bX}) $\to$ simplificar a una FGM conocida $\to$ identificar y concluir por unicidad.

4. Gamma y Beta (truco del núcleo)

$$\Gamma(\alpha)=\int_0^\infty x^{\alpha-1}e^{-x}dx,\quad \alpha>0;$$

$$\Gamma(\alpha+1)=\alpha\Gamma(\alpha),\quad \Gamma(n)=(n-1)!,\quad \Gamma(\tfrac{1}{2})=\sqrt{\pi}.$$

$$B(\alpha,\beta)=\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Núcleo: $\int_0^\infty x^{\alpha-1}e^{-x/\beta}dx=\beta^\alpha\Gamma(\alpha)$, $\int_0^1x^{\alpha-1}(1-x)^{\beta-1}dx=B(\alpha,\beta)$, $\int e^{-\frac{(x-\mu)^2}{2\sigma^2}}dx=\sqrt{2\pi\sigma^2}$. **Momentos Gamma:** $\mathbb{E}(X^k)=\beta^k\frac{\Gamma(\alpha+k)}{\Gamma(\alpha)}$.

5. Distribuciones conjuntas

$$f_X(x)=\int f_{XY}(x,y)\,dy,\quad f_{X|Y}(x\mid y)=\frac{f_{XY}(x,y)}{f_Y(y)},$$

$$X\perp\!\!\!\perp Y\iff f_{XY}=f_Xf_Y.$$

$$\mathbb{E}[g(X,Y)]=\iint g\,f_{XY}\,dx\,dy,\quad \mathbb{E}[X\mid Y=y]=\int x f_{X|Y}dx,$$

$$\mathbb{E}[\mathbb{E}(X\mid Y)]=\mathbb{E}X.$$

Discretas: cambia $\int$ por $\sum$.

- Soporte **no rectangular** (p. ej. $0<x<y<1$) $\Rightarrow$ no son independientes.
- Soporte rectangular y $f_{XY}=g(x)h(y)\Rightarrow$ sí independientes.
- **Límites:** dibuja el soporte; la variable de afuera tiene límites constantes; la de adentro, fija la de afuera y mira dónde entra/sale; si la frontera cambia, parte la integral.

6. Transformaciones (Clase 1)

Acumulada (general): (1) soporte de $U=h(Y)$; (2) $F_U(u)=\mathbb{P}(h(Y)\le u)$; (3) despeja Y (si h decrece, voltea); (4) usa F_Y; (5) $f_U=F_U'$.

$$\text{Jacobiano 1D (monótona):}\quad f_U(u)=f_Y[h^{-1}(u)]\Big|\frac{dh^{-1}}{du}\Big|.$$

$$\text{No monótona:}\quad f_U(u)=\sum_k f_Y[h_k^{-1}(u)]\Big|\frac{dh_k^{-1}}{du}\Big|,$$

$$U=Y^2:\quad f_U(u)=\frac{f_Y(\sqrt{u})+f_Y(-\sqrt{u})}{2\sqrt{u}}\ (u>0).$$

$$\text{Bivariado:}\quad f_{U_1,U_2}=f_{Y_1,Y_2}(h_1^{-1},h_2^{-1})\,|J|,\quad J=\det\big[\partial y_i/\partial u_j\big];$$

$$n\text{-D:}\quad g(\mathbf{u}) = f(\mathbf{y}(\mathbf{u}))\left|\det\frac{\partial \mathbf{y}}{\partial \mathbf{u}}\right|.$$

J es el jacobiano de la **inversa**; atajo $J = 1/\det(\partial(u_1,u_2)/\partial(y_1,y_2))$. Si piden una sola U_1, añade una **auxiliar** y marginaliza. Tramos planos $\Rightarrow$ átomos: usa sólo la acumulada.

7. Estadísticos de orden (Clase 2)

$Y_{(1)} = \text{mín}$, $Y_{(n)} = \text{máx}$ de Y_i i.i.d.

$$F_{(n)}(y) = [F(y)]^n,\quad g_{(n)}(y) = n[F(y)]^{n-1}f(y);$$
$$F_{(1)}(y) = 1-[1-F(y)]^n,\quad g_{(1)}(y) = n[1-F(y)]^{n-1}f(y).$$

$$\text{Teo. 1:}\quad f_{Y_{(i)}}(x) = \frac{n!}{(i-1)!(n-i)!}[F(x)]^{i-1}f(x)[1-F(x)]^{n-i}.$$

$$f_{(1),\dots,(n)} = n!\prod_k f(y_k);$$

$$f_{(i),(j)}(x,y) = c_{ij}\,f(x)f(y)\,F(x)^{i-1}[F(y)-F(x)]^{j-i-1}[1-F(y)]^{n-j},$$
$$c_{ij} = \frac{n!}{(i-1)!(j-i-1)!(n-j)!}\ (i<j).$$

Mín. de $\text{Exp}(\beta)$ i.i.d. $\sim \text{Exp}(\beta/n)$; $Y_{(i)}$ de $U(0,1) \sim \text{Beta}(i, n-i+1)$.

8. Distribuciones muestrales: χ^2 y t (Clase 3)

$Y_1,\dots,Y_n$ i.i.d. $N(\mu,\sigma^2)$; $\bar{Y} = \frac{1}{n}\sum Y_i$, $S^2 = \frac{1}{n-1}\sum(Y_i-\bar{Y})^2$.

$$\bar{Y} \sim N\Big(\mu, \frac{\sigma^2}{n}\Big) \;\Leftrightarrow\; Z = \frac{\bar{Y}-\mu}{\sigma/\sqrt{n}} \sim N(0,1);$$
$$\sum_i \Big(\frac{Y_i-\mu}{\sigma}\Big)^2 \sim \chi^2_{(n)}.$$

$$\bar{Y} \perp\!\!\!\perp S^2,\quad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)};$$
$$\chi^2_{(\nu)} = G(\tfrac{\nu}{2},2):\ \mathbb{E}=\nu,\ \text{Var}=2\nu,\ M=(1-2t)^{-\nu/2}.$$

$$Z\sim N(0,1) \perp\!\!\!\perp W\sim\chi^2_{(\nu)} \Rightarrow T = \frac{Z}{\sqrt{W/\nu}} \sim t_{(\nu)};$$
$$\nu = n-1:\quad \frac{\bar{Y}-\mu}{S/\sqrt{n}} \sim t_{(n-1)}.$$

$t_{(\nu)}$: $\mathbb{E}=0\,(\nu>1)$, $\text{Var}=\frac{\nu}{\nu-2}\,(\nu>2)$, $\to N(0,1)$ si $\nu\to\infty$. FDA: $F_t(t) = 1-\frac{1}{2}I_{\nu/(\nu+t^2)}(\frac{\nu}{2},\frac{1}{2})$ $(t\ge 0)$.

9. La distribución F (Clase 4)

$$W_1\sim\chi^2_{(\nu_1)} \perp\!\!\!\perp W_2\sim\chi^2_{(\nu_2)} \Rightarrow F = \frac{W_1/\nu_1}{W_2/\nu_2} \sim F_{(\nu_1,\nu_2)};$$
$$\mathbb{E} = \frac{\nu_2}{\nu_2-2}\,(\nu_2>2),\quad \text{Var} = \frac{2\nu_2^2(\nu_1+\nu_2-2)}{\nu_1(\nu_2-2)^2(\nu_2-4)}\,(\nu_2>4).$$

FDA: $F(x) = I_{\nu_1 x/(\nu_1 x+\nu_2)}\big(\frac{\nu_1}{2},\frac{\nu_2}{2}\big)$; I = beta incompleta regularizada.

- $X\sim t_{(\nu)} \Rightarrow X^2\sim F_{(1,\nu)}$, y $F_{(1,\nu,\alpha)} = t^2_{(\nu),\alpha/2}$.
- $X\sim F_{(\nu_1,\nu_2)} \Rightarrow 1/X\sim F_{(\nu_2,\nu_1)}$; $F_{(\nu_1,\nu_2,1-\alpha)} = 1/F_{(\nu_2,\nu_1,\alpha)}$.
- Muestras normales indep.: $\dfrac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{(n_1-1,n_2-1)}$; si $\sigma_1^2=\sigma_2^2$, $S_1^2/S_2^2\sim F_{(n_1-1,n_2-1)}$.

10. Teorema del límite central (Clase 5)

Y_i i.i.d., $\mathbb{E}Y_i=\mu$, $\text{Var}=\sigma^2<\infty$; $S_n=\sum Y_i$:

$$W_n = \frac{\bar{Y}-\mu}{\sigma/\sqrt{n}} = \frac{S_n-n\mu}{\sigma\sqrt{n}} \xrightarrow{\ d\ } N(0,1);$$
$$\bar{Y}\overset{a}{\sim} N\Big(\mu,\frac{\sigma^2}{n}\Big),\quad S_n\overset{a}{\sim} N(n\mu,n\sigma^2).$$

Bernoulli: $\text{Bin}(n,p)\overset{a}{\sim} N(np, np(1-p))$ si $np\ge 5$ y $n(1-p)\ge 5$. Sin σ^2 finita no hay TLC.

- **Corrección $\pm\frac{1}{2}$:** $X\le k \Rightarrow \Phi(\frac{k+0.5-\mu_X}{\sigma_X})$; $X<k\Rightarrow\Phi(\frac{k-0.5-\mu_X}{\sigma_X})$; $X\ge k\Rightarrow 1-\Phi(\frac{k-0.5-\mu_X}{\sigma_X})$; $X>k\Rightarrow 1-\Phi(\frac{k+0.5-\mu_X}{\sigma_X})$.
- $X=k \Rightarrow \Phi(\frac{k+0.5-\mu_X}{\sigma_X})-\Phi(\frac{k-0.5-\mu_X}{\sigma_X})$; $a\le X\le b\Rightarrow \Phi(\frac{b+0.5-\mu_X}{\sigma_X})-\Phi(\frac{a-0.5-\mu_X}{\sigma_X})$.

11. Tablas y valores críticos

Las tablas ESPOL dan área a la **izquierda** q: $\mathbb{P}(X\le x)=q$. **Crítico con cola derecha α = tabla en columna $q=1-\alpha$.** El percentil $100p$ es x_p; el crítico al 5 % es el percentil 95.

- Z: $\Phi(z_\alpha)=1-\alpha$, $\Phi(-z)=1-\Phi(z)$.
- $\chi^2_{(\nu),\alpha}$: columna $1-\alpha$. $t_{(\nu),\alpha}$: columna $1-\alpha$; $\mathbb{P}(|T|\le c) = 2\mathbb{P}(T\le c)-1$ (simetría).
- $F_{(\nu_1,\nu_2)}$: bloque ν_2 (denominador), $F_{(\nu_1,\nu_2,\alpha)}$ = fila $q=1-\alpha$, columna ν_1 (numerador).